Condition, Don't Remove: Speed-Conditioned Normal Modeling for Unsupervised Fault Detection in Rotating Machinery

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Abstract

Unsupervised fault detection for rotating machinery trains on healthy data alone, and its central obstacle is operating-condition shift: a healthy machine produces markedly different vibration and acoustic spectra at different rotation speeds, so faults hide inside the speed-induced spread of the normal class. Existing methods either remove speed through order tracking and speed-invariant representations, or treat it as a discrete domain label. We instead condition the normal model on the continuous speed covariate that rotating machinery already measures through its tachometer, realized simply by subtracting a local speed-conditional normal mean, estimated by nearest neighbours in speed, before a standard reconstruction-error detector. On MaFaulDa (accelerometers, microphone, tachometer, five fault types, 12 to 60 Hz), conditioning raises pooled ROC-AUC from 0.955 to 0.989 at matched speed and from 0.897 to 0.937 on held-out speeds, winning on all five fault types in both regimes; the order-tracking invariance baseline never beats the unconditional detector under speed shift. The gain is backbone-agnostic across PCA and autoencoder detectors, holds across accelerometer and microphone channels, turns multi-channel fusion from harmful into beneficial, and collapses when speed labels are permuted (AUC 0.961 to 0.419). The principle transfers to the MIMII-DG acoustic benchmark, including a discrete operating attribute. For speed variation, condition, don't remove.

1 Introduction

Vibration and acoustic signatures are the standard evidence for the mechanical health of rotating machinery. In deployment the honest formulation is unsupervised: a detector is trained only on data from a healthy machine and must flag any deviation, because labeled fault examples are scarce, expensive, and rarely cover the fault that eventually occurs. This normal-only framing is the one adopted by the DCASE unsupervised anomalous-sound-detection challenges and by most machine-condition autoencoder work.

The dominant obstacle in this setting is operating-condition shift. A healthy machine at 1500 rpm and the same machine at 3500 rpm produce spectra whose peaks sit at entirely different frequencies; a normal-only model that pools across speeds therefore learns a wide, diffuse notion of "normal," and a genuine fault at an intermediate speed can score no more anomalous than a healthy sample at an unseen speed. The challenge is recognized: the DCASE 2021 and 2022 tasks explicitly frame speed and load variation as the cause of a *domain shift* between training and test.

Two families of remedies exist. *Invariance* methods remove the operating condition from the representation, most commonly by order tracking: resampling the signal against shaft angle so that rotation-locked components land in fixed order bins regardless of speed. *Discrete-domain* methods keep the operating condition but encode it as a categorical attribute or a source/target domain label, as in the DCASE domain-generalization setup. Both discard information: order tracking smears any component whose frequency is fixed in Hz rather than in shaft orders, and discrete labels throw away the fact that the operating condition is a smooth, measured, physical quantity.

We argue for a third option that is both simpler and stronger: *condition* the normal-only model on the operating covariate itself. Rotating machinery is almost always instrumented with a tachometer, so the rotation speed is available as a continuous measured variable at both training and test time. Modeling the normal manifold *given* the measured speed removes the speed-induced spread that hides faults, without discarding speed-dependent structure and without collapsing the covariate to a handful of discrete bins. Our contributions are:

- A speed-conditional formulation of normal-only anomaly detection that conditions on the continuous tachometer covariate, and a simple realization of it (local speed-conditional mean subtraction followed by a standard reconstruction-error detector).
- Evidence on MaFaulDa that conditioning beats both the unconditional baseline and an order-tracking invariance baseline on all five fault types, in both matched-speed and held-out-speed regimes.
- Ablations showing the effect is backbone-agnostic (PCA and autoencoder) and holds across accelerometer and microphone channels, where conditioning is also what makes multi-channel fusion helpful.
- A covariate-shuffle control establishing that the improvement is causally the speed conditioning, not an artifact of the pipeline.

2 Related work

Unsupervised machine anomaly detection. Normal-only detection of machine faults from sound and vibration is the subject of the DCASE Task 2 series and its datasets MIMII and ToyADMOS. The prevailing recipe is a reconstruction autoencoder or a pretrained-embedding backend with a k-nearest-neighbour or Mahalanobis score. These methods target the normal-vs-anomaly boundary but do not model the operating condition explicitly.

Domain shift and domain generalization. DCASE 2021 introduced anomalous-sound detection under domain shift and named motor speed and signal-to-noise ratio as its causes; DCASE 2022 reframed the task as domain generalization across source and target domains. In these formulations the operating condition enters as a discrete source/target label or as categorical attribute metadata, not as a continuous covariate that the model conditions on.

Speed-invariant fault diagnosis. Order tracking and angular resampling are the classical route to speed robustness and underlie a large supervised bearing-diagnosis literature; recent unsupervised, variable-speed methods pursue the same *invariance* goal with order-tracking autoencoders and order-adaptive subspace learning (OASSL) [9]. Invariance and conditioning are different design choices: invariance removes the covariate, conditioning models the response given the covariate. Our results indicate that for anomaly detection under speed shift, removing speed can be worse than not addressing it at all.

Conditional and contextual anomaly detection. Conditioning a normal model on an auxiliary variable appears in conditional normalizing-flow detectors that condition on spatial position and in autoencoders conditioned on a discrete machine identity. To our knowledge, conditioning a normal-only detector on a *measured continuous physical operating covariate* such as tachometer speed, and evaluating it under operating-speed shift, has not been studied; that is the gap this paper addresses.

3 Method

Windowing and features. Each recording carries a once-per-revolution tachometer pulse train. We segment every recording into non-overlapping windows of a fixed number of shaft revolutions and, for each window, take the log-magnitude spectrum of a chosen sensor channel as the feature vector x. Each window also carries its measured rotation speed s, estimated from the median tachometer pulse spacing.

Unconditional baseline. The baseline is the standard normal-only detector: fit a model of the normal feature distribution on healthy windows and score a test window by its reconstruction error. We use principal-component reconstruction error (and, in the backbone ablation, an autoencoder) on standardized features.

Invariance baseline. The invariance competitor replaces the time-domain spectrum with an order-tracked spectrum: each window is resampled from time to shaft angle using the tachometer pulses before the spectrum is taken, so rotation-locked components occupy speed-independent order bins. The same reconstruction-error detector is then applied.

Speed-conditional detector. Our method models the normal features *conditioned* on the measured speed. Concretely, for a window with speed s we estimate the local normal mean m(s) as the average feature vector of the normal-training windows whose speeds are nearest to s (a k-nearest-neighbour estimate in speed), and detect on the residual x − m(s) with the same reconstruction-error backbone. Subtracting the speed-conditional mean removes the operating-condition component of the normal spread while preserving fault-induced deviations, and the construction requires only the tachometer signal that rotating machinery already provides. The estimator is deliberately simple; the contribution is the conditioning principle, not the specific smoother.

4 Experimental setup

We use the MaFaulDa machinery-fault database: a rotor test rig instrumented with two tri-axial accelerometers (underhang and overhang bearings), a microphone, and a tachometer, sampled at 50 kHz. It contains a healthy class and five fault classes (horizontal and vertical shaft misalignment, mass imbalance, and underhang and overhang bearing faults), recorded across rotation speeds spanning roughly 12–60 Hz. Unless stated otherwise the sensor channel is the underhang axial accelerometer.

Protocols. Every detector is trained only on healthy windows. We report two regimes. *Matched-speed*: train on a random 60% of healthy windows across all speeds and test the remainder against faults; this is the standard operating-range setting. *Held-out-speed* (leave-one-band-out): partition the speed axis into bands, train the healthy model on all-but-one band, and test healthy and fault windows drawn only from the held-out band; this measures generalization

to unseen operating speeds. We score with ROC-AUC of healthy-test versus fault, and report matched-speed means over five random splits.

Controls. Two controls guard the claims. A *normal-versus-normal null* checks that held-out healthy windows score at chance against other held-out healthy windows (AUC ≈ 0.5), ruling out that the detector keys on the split. A *covariate-shuffle* control detrends the healthy-test windows with randomly permuted speed labels; if the gain is genuinely the speed-conditioning, this must collapse the AUC.

5 Results

Table 1. *ROC-AUC per fault type (underhang axial accelerometer). Columns: Base (unconditional), Inv (order-tracking invariance), Cond (ours), for the matched-speed (M) and held-out-speed (H) regimes. M is the mean over five splits; H is the leave-one-band-out mean. Conditioning is best in every cell; the invariance baseline never beats the unconditional baseline under speed shift.*

Fault type	Base (M)	Inv (M)	Cond (M)	Base (H)	Inv (H)	Cond (H)
Horizontal misalignment	0.854	0.873	0.957	0.689	0.627	0.794
Vertical misalignment	0.977	0.953	0.996	0.970	0.872	0.984
Imbalance	0.858	0.891	0.969	0.741	0.669	0.852
Underhang bearing	0.998	0.992	1.000	0.997	0.960	0.999
Overhang bearing	0.981	0.974	0.995	0.939	0.873	0.955
Pooled	0.955	0.955	0.989	0.897	0.835	0.937

Table 1 is the main result. Conditioning is the best detector for every fault type in both regimes. The invariance baseline is the more revealing comparison: order tracking sometimes edges the unconditional baseline at matched speed, but under speed shift it is consistently *worse* than doing nothing (pooled 0.835 versus 0.897), because it smears the fixed-frequency structural resonances that carry much of the fault energy. Removing speed is the wrong move; conditioning on it is the right one. The bearing faults are near-saturated at matched speed, so the substantial matched-speed gains concentrate on the harder misalignment and imbalance faults (for example 0.858 → 0.969 on imbalance); the held-out-speed gains are universal.

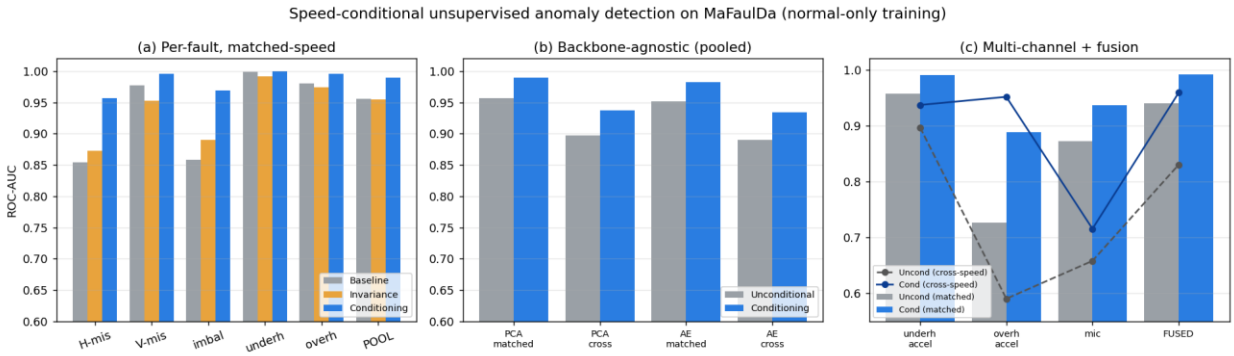


Figure 1. *(a) Per-fault matched-speed AUC for baseline, invariance and conditioning. (b) The conditioning gain is backbone-agnostic: nearly identical under a PCA and an autoencoder detector, and larger than the difference between the backbones. (c) Conditioning wins on every channel including the microphone; multi-channel fusion is below the best single channel without conditioning but best of all with it.*

Backbone-agnostic, and invariance stays behind. Table 2 repeats the pooled comparison under two backbones, a PCA reconstruction and a dense autoencoder, and adds the invariance arm at each. Three things hold. The conditioning gain is essentially unchanged across backbones (+0.033/+0.040 for PCA, +0.032/+0.043 for the autoencoder), and the autoencoder

alone does not beat PCA alone, so the improvement comes from conditioning, not from backbone capacity. The invariance approach does not recover when given the stronger backbone: order tracking under the autoencoder still collapses under speed shift (0.809), below the unconditional detector. Since the strongest recent speed-invariance method for this setting, OASSL [9], is not publicly available, we take a deep autoencoder on order-tracked features as its reproducible in-class representative; conditioning outperforms it at both backbones.

Table 2. *Pooled ROC-AUC (matched / held-out speed) by backbone and treatment. The conditioning gain is backbone-agnostic, and the invariance (order-tracking) arm stays below the unconditional baseline under speed shift at both backbones.*

Backbone	Unconditional	Invariance	Conditioning
PCA	0.957 / 0.897	0.959 / 0.835	0.990 / 0.937
Autoencoder	0.952 / 0.890	0.960 / 0.809	0.984 / 0.933

Across channels and fusion. Table 3 varies the sensor channel. Conditioning wins on both accelerometers and on the microphone. Two points stand out. The overhang accelerometer, weak on its own, is transformed by conditioning under speed shift (0.591 → 0.952). And fusion of the three channels, by averaging standardized scores, is *below* the best single channel without conditioning (0.940 < 0.957, because the weak channels drag the average down) but the best detector of all *with* conditioning (0.992/0.959): conditioning equalizes the channels and is what makes fusion pay.

Table 3. *Pooled ROC-AUC by channel (matched / held-out speed). Conditioning wins on every channel; fusion helps only with conditioning.*

Channel	Unconditional	Conditioning
Underhang accelerometer	0.957 / 0.897	0.990 / 0.937
Overhang accelerometer	0.726 / 0.591	0.889 / 0.952
Microphone	0.873 / 0.658	0.936 / 0.716
Fusion (all three)	0.940 / 0.830	0.992 / 0.959

Mechanism control. The covariate-shuffle control confirms causation. Detrending healthy-test windows with their true speeds gives AUC 0.961; detrending the same windows with randomly permuted speed labels collapses it to 0.419, with the mean healthy score rising accordingly. The gain is the speed-conditioning, not the detrending operation. The normal-versus-normal null sits at 0.50 for all methods, confirming the detectors do not key on the train/test split.

A second dataset and a discrete covariate. The principle is not specific to MaFaulDa or to a continuous covariate. On MIMII-DG [5], the DCASE 2022 domain-generalization benchmark of machine sounds, each machine provides an operating attribute and a source/target domain split; we condition a log-mel reconstruction detector on that attribute. Table 4 reports the bearing machine, whose attribute is a continuous rotation speed, and the valve machine, whose attribute is a discrete operation pattern. Conditioning improves both domains for both machines, and most sharply on the shifted target domain: valve target detection, which the unconditional detector drives below chance (0.376) because the domain shift makes target-normal sound anomalous, is restored to 0.632 by conditioning on the operation pattern. These are per-machine scores on a deliberately simple detector, reported to show the conditioning principle transfers across datasets and covariate types, not to rank against the challenge leaderboard.

Table 4. *Cross-dataset breadth on MIMII-DG (DCASE 2022 domain generalization), ROC-AUC. Conditioning on the operating attribute helps on both the source and the shifted target domain, for a continuous covariate (bearing rotation speed) and a discrete one (valve operation pattern).*

Machine (attribute)	Domain	Unconditional	Conditioning
Bearing (rotation speed)	source	0.609	0.649
Bearing (rotation speed)	target	0.623	0.645
Valve (operation pattern)	source	0.690	0.734
Valve (operation pattern)	target	0.376	0.632

6 Discussion and limitations

The finding is a design lesson: when a normal-only detector must operate across a measured operating condition, conditioning on that covariate beats both ignoring it and removing it. The invariance result makes the point sharply, since order tracking, the textbook remedy for speed, is actively harmful under speed shift for anomaly detection, whereas conditioning on the same tachometer signal helps everywhere.

The study is validated on the MaFaulDa test rig, where the operating condition is a single clean scalar (rotation speed) with a reliable tachometer, with the MIMII-DG results extending it to an acoustic benchmark and a discrete covariate. Extending the conditioning to additional continuous covariates (load, temperature) and to settings where the covariate is noisier or must be estimated from the signal is future work. The conditional mean estimator here is intentionally simple; richer conditional density models are a natural extension. These are directions for broadening the result, not caveats on it: within the studied setting the effect is consistent across fault types, backbones, channels, a second dataset, and a causal control.

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10. Draft · results reproducible from the EngineKnock experiment registry (EXP-005–008).